

RhizoChip // Technical Documentation

Version: 0.1

Last updated: May 2026

Contributors: RhizoChip is developed by the **Systems Biology Program** at the **Center for Genomic Sciences, UNAM**, as part of an effort to build reproducible, open, and modular experimental pipelines for microbial ecology and experimental evolution.

Development: Carles Tardío Pi (tardiopi@ccg.unam.mx), Rafael Peña Miller (rpm@ccg.unam.mx)

Experimental users and contributors: Nicole Gomez, Gustavo Cuaxinque, Andrés López, Esperanza Martinez, Ayari Fuentes

License: RhizoChip is intended as an open research tool. Unless stated otherwise, the source code is distributed under the MIT License. The documentation, diagrams, and written material are distributed under a Creative Commons Attribution 4.0 International License (CC BY 4.0). Users may use, modify, and redistribute the code and documentation, provided that appropriate credit is given to the original contributors.

1. Overview

This device is a modular experimental platform designed for the controlled co-culture of microbial communities under laboratory and environmentally relevant conditions. The system enables the spatial confinement of microbial populations while allowing the exchange of diffusible metabolites and chemical signals through permeable membranes. It is intended for the study of microbial interactions, community assembly, and functional performance of synthetic microbial consortia, with a particular focus on soil and plant-associated microbiomes.

The platform is being developed as a tool to support the identification and optimization of microbial consortia with potential applications in sustainable agriculture, including biofertilization, plant growth promotion, and soil health improvement.

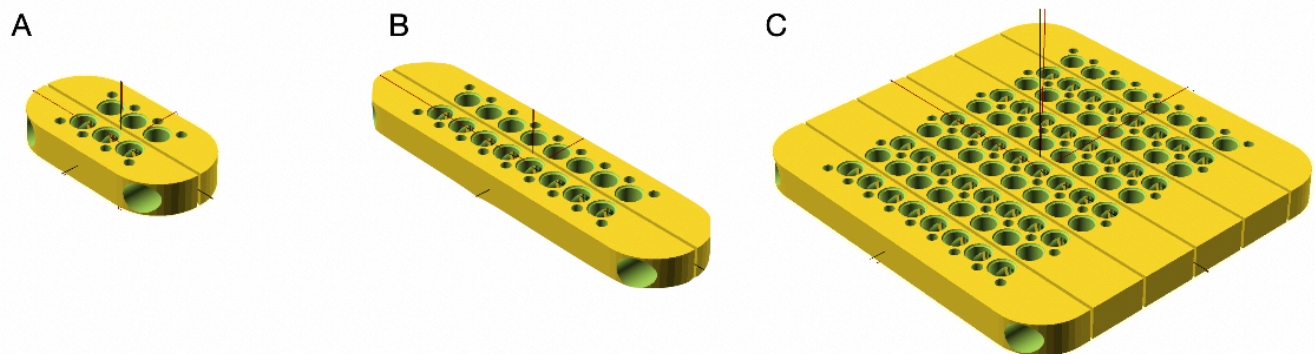


Figure 1. Modular architecture of the microbial co-culture device.

(A) 3D render of a miniaturized configuration with two modules and three rows of paired compartments.

(B) 3D render of a two-module configuration containing eight pairs of connected wells, designed for parallel co-culture assays.

(C) 3D render of an expanded modular configuration composed of five paired module sets, each containing eight rows, illustrating the scalability of the device for larger experimental designs.

2. Design Principle

The device consists of modular rows of compartments that can be connected in paired configurations (**Figure 1**). Each compartment is designed to contain a microbial isolate or a defined microbial community. Adjacent compartments are separated by permeable membranes, allowing the diffusion of metabolites and chemical signals while preventing direct physical mixing of populations.

This architecture enables controlled comparison between connected and isolated configurations. In this way, the system can distinguish interaction-mediated effects from autonomous growth dynamics.

The design is implemented as a parametric OpenSCAD model, which allows the geometry of the device to be modified directly from code. Key parameters include the number of modules, number of rows and columns, module dimensions, well diameter, well spacing, lateral ports, screw holes, rod channels, lid geometry, and label placement. This makes the platform adaptable to different experimental scales, from small interaction assays to larger multi-module screens.

The modular design supports several experimental configurations, including:

- paired compartments for testing direct interaction effects
- multiple connected rows for higher-throughput interaction screening
- isolated rows as non-interaction controls
- expanded multi-block systems for larger community designs
- separate body and lid components for assembly, containment, and handling

3. Fabrication and Digital Design Workflow

The device is generated from a parametric OpenSCAD script. The code produces the 3D geometry of the device body and lid, and allows users to select between different configurations by editing the parameter block at the beginning of the file. For example, the user can define the number of modules, the number of rows, the number of columns, and the overall module dimensions.

The script includes predefined configurations, such as compact two-module devices, larger two-module devices with multiple rows, and multi-module layouts. The user can also define custom geometries by modifying the relevant parameters. The model includes integrated features for wells, lateral ports, vertical screw holes, horizontal rod channels, rounded module edges, and engraved labels.

A typical workflow is:

1. Open the `.scad` file in OpenSCAD.
2. Select whether to render the body or the lid using the `render_body` and `render_lid` variables.
3. Choose or modify the desired device configuration by setting parameters such as `N`, `nrows`, `ncols`, `module_length`, and `module_width`.
4. Preview and render the model.
5. Export the resulting geometry as an STL file.
6. Slice the STL file using standard 3D-printing software.
7. Fabricate the component using extrusion-based 3D printing.

Initial prototypes have been fabricated using 3D printing, primarily through extrusion-based additive manufacturing (**Figure 2**). This approach enables rapid prototyping, iterative design modifications, and low-cost production of custom geometries.

The current designs include devices fabricated in PLA and PC. Future optimization will focus on material compatibility with soil conditions, mechanical stability, sterilization procedures, membrane integration, and reproducibility across printing batches. The same digital design workflow can be adapted to other materials or manufacturing methods depending on the intended application and required robustness.

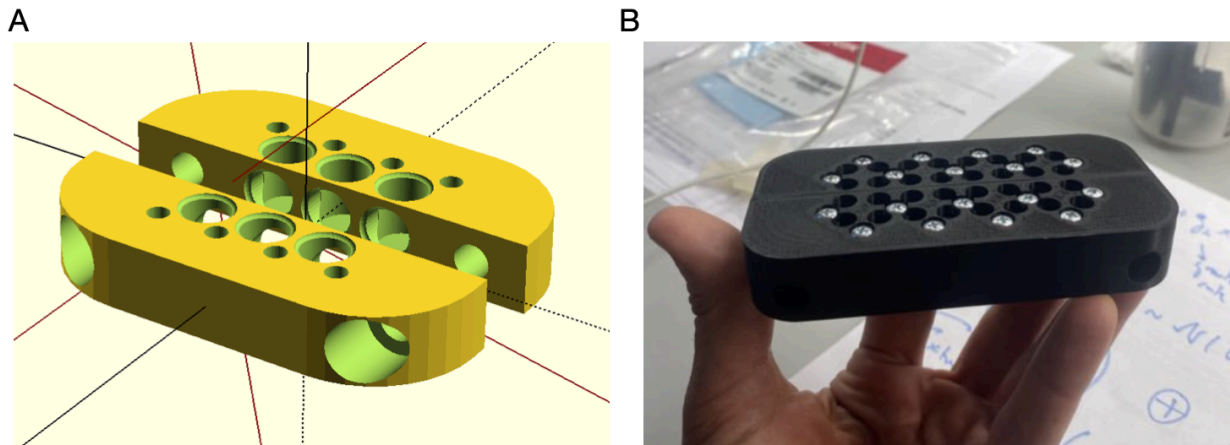


Figure 2. Design and fabrication of the microbial co-culture device.

(A) 3D render of a module with three compartments connected in paired configurations, allowing metabolite exchange between adjacent units. (B) Prototype of the device fabricated by 3D printing.

4. Experimental Use

The device is designed for use in several experimental contexts:

1. **Laboratory experiments**

The system can be inoculated with defined microbial communities and incubated under controlled conditions to evaluate containment, reproducibility, and metabolite-mediated interactions.

2. **Controlled soil or mesocosm assays**

The device can be placed in soil-like environments, compost, or pot systems to assess microbial growth and interaction dynamics under more complex physicochemical conditions.

3. **In situ agricultural testing**

The device can be deployed directly in agricultural soils to evaluate microbial interactions under field-relevant conditions.

In ongoing pilot work, the device has been used with synthetic microbial communities derived from bean root-associated microbiomes. These communities are being evaluated for their potential to assemble, interact, and support functions relevant to plant productivity.

5. Functional Validation

The device has been evaluated through a series of validation steps designed to test its performance under controlled experimental conditions. These include:

- confirmation of physical containment between compartments
- assessment of membrane-mediated exchange
- evaluation of microbial growth inside the device
- comparison between connected and isolated configurations
- reproducibility across device replicates
- recovery of microbial samples after incubation
- functional testing of assembled consortia in plant-associated assays

Physical containment was evaluated through contamination-control experiments in which the device was exposed to external bacterial cultures while monitoring whether the internal compartments remained uncontaminated. These assays confirmed that the device maintains physical separation between populations while allowing controlled experimental handling (**Figure 3**).

Microbial growth and recovery were tested using defined microbial communities, including synthetic communities derived from bean root-associated microbiota. These experiments demonstrated that the device supports microbial incubation and downstream recovery of samples for analysis.

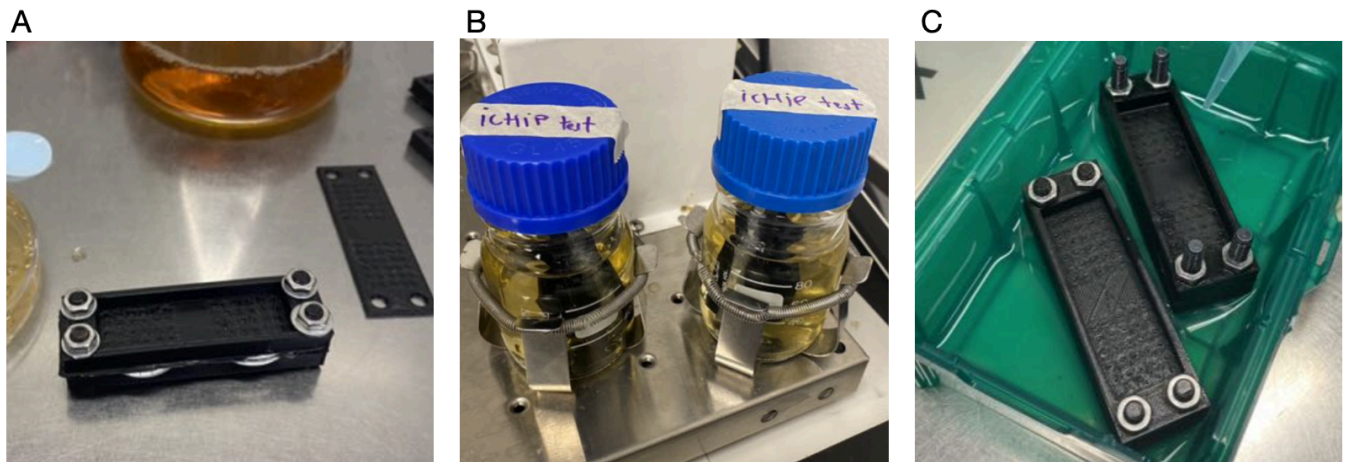


Figure 3. Validation of the device under controlled laboratory conditions.

(A) Co-culture device inoculated with a synthetic community derived from bean roots, composed of seven strains. (B) Co-culture device inoculated with the same seven-strain synthetic community, used to evaluate system performance under controlled conditions. (C) Containment test in which the device was exposed to culture medium containing external bacteria. No internal contamination was observed, demonstrating physical isolation between compartments.

Functional validation is currently in progress through plant-associated assays in which assembled microbial communities are being evaluated for their effects on relevant plant phenotypes, including nodulation and growth-associated responses. This ongoing work is intended to assess the capacity of the platform to link microbial community structure and interaction patterns with functional outcomes in plant-associated systems.

6. Analytical Framework

The device generates quantitative data that can be used to infer microbial interactions. Growth, abundance, community composition, and functional readouts can be compared across experimental configurations.

Data analysis may include:

- comparison of growth in connected versus isolated compartments,
- inference of positive, negative, or neutral interactions,
- identification of stable consortia,
- detection of community-level functional effects,
- integration with ecological models or network inference methods.

The device is therefore not only a culture platform, but also part of an integrated experimental workflow for studying microbial interaction networks.

7. Technological Relevance

The platform addresses a current limitation in microbial ecology and agricultural biotechnology: the lack of experimental systems that connect controlled co-culture assays with environmentally relevant soil conditions. Existing in situ cultivation tools, such as iChip-like devices, have been useful for recovering previously uncultivable microorganisms, but are not primarily designed to quantify interaction networks among defined microbial communities.

This device aims to bridge that gap by combining:

- spatial confinement,
- controlled metabolite exchange,
- modular experimental design,
- compatibility with synthetic communities,
- potential deployment in soil or agricultural environments.

8. Development Stage

The current state of development corresponds to an intermediate technology readiness level, between TRL 4 and TRL 6. The system has been designed, fabricated, and tested as a functional prototype under laboratory conditions, demonstrating the basic operation of the co-culture architecture and the integration of its main components.

Current work focuses on advancing the platform from laboratory validation toward demonstration in relevant agricultural environments. This includes improving material selection, reproducibility, structural robustness, membrane integration, and performance under soil-associated conditions.

The next development phase will support the transition from TRL 4, corresponding to validation of an integrated prototype in the laboratory, toward TRL 6, corresponding to demonstration of the system in relevant agricultural settings. This will involve testing the device in controlled soil systems, mesocosms, and field-like conditions to evaluate its performance, reliability, and capacity to generate reproducible data on microbial interactions in soil.

9. Intended Applications

Potential applications include:

- identification of microbial consortia for sustainable agriculture,
 - screening of plant-associated microbial communities,
 - evaluation of biofertilizer or bioprotective formulations,
 - study of microbial interaction networks in soil,
 - development of microbiome-based agricultural technologies,
 - experimental validation of synthetic microbial communities.
-

10. Conclusion

The in situ microbial co-culture device provides a modular and adaptable platform for studying microbial interactions under controlled and environmentally relevant conditions. By enabling spatial confinement and metabolite exchange, the system supports the quantitative analysis of microbial interactions and the identification of functional consortia.

Its development has potential relevance for agricultural biotechnology, particularly for the rational design and validation of microbial consortia aimed at improving productivity and sustainability in agricultural systems.